

DESIGN TASK

1. A healthcare organization has collected patient waiting-time data from 20 hospitals over one year.

Task: Design a visualization dashboard that:

- Shows **Box Plots** and **Violin Plots** for waiting times across hospitals.
- Identifies outliers and extreme waiting-time cases.
- Highlights differences in median waiting times and variability among hospitals.

Objective: Help hospital administrators identify service inefficiencies and improve patient care quality.

Aim

To design an interactive Tableau dashboard that analyzes patient waiting times across 20 hospitals using **Box Plots** and **Violin Plots** to identify outliers, compare median waiting times, and analyze variability in hospital service efficiency.

Procedure

1. Import the Dataset

1. Open **Tableau Desktop/Public**.
2. Click **Connect** → **Text File** and select **Hospital_Waiting_Time.csv**.
3. Verify the fields and open **Sheet 1**.

2. Create a Box Plot

1. Drag **Hospital** to the **Columns** shelf.
2. Drag **Waiting_Time** to the **Rows** shelf.
3. Click **Show Me** and select **Box-and-Whisker Plot**.
4. Rename the worksheet as **Box Plot**.

3. Create a Violin Plot

Note: Tableau does **not** provide a built-in Violin Plot.

1. Import a dataset that contains the required density values (or use a Tableau extension if available).
2. Drag **Hospital** to **Columns**.
3. Drag **Waiting_Time** to **Rows**.
4. Create the Violin Plot using the density field or extension and rename the worksheet as **Violin Plot**.

4. Create the Dashboard

1. Click **New Dashboard**.
2. Add the **Box Plot** and **Violin Plot** worksheets.
3. Add a **Hospital** filter and apply it to all worksheets.
4. Save the dashboard.

Explanation

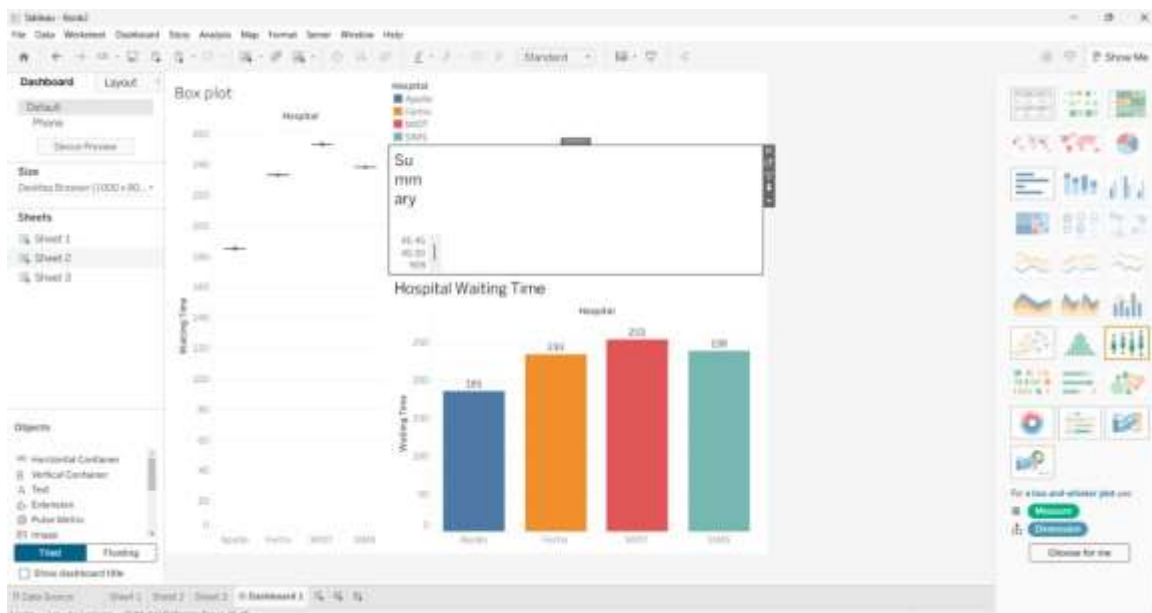
Box Plot

A box plot summarizes the distribution of patient waiting times for each hospital. It displays the median, quartiles, variability, and outliers. This helps identify hospitals with unusually long waiting times and compare overall service performance.

Violin Plot

A violin plot combines a box plot with a density plot to show both the distribution and concentration of waiting times. The width of the plot indicates how frequently particular waiting times occur. It helps compare waiting-time patterns and variability among hospitals.

Output



Result

The Tableau dashboard successfully displays patient waiting-time distributions using Box Plots and Violin Plots. It helps identify outliers, compare median waiting times, analyze variability among hospitals, and supports administrators in improving patient care quality.

2. A smart city project monitors hourly air quality data (PM2.5, PM10, NO₂, and CO) from 15 monitoring stations.

Task: Create a visualization design that:

- **Uses Heatmaps to display pollution intensity across locations and time periods.**
- **Employs Ridgeline Plots to compare pollutant distributions among stations.**
- **Highlights pollution spikes and abnormal environmental conditions.**

Objective: Assist environmental authorities in identifying pollution hotspots and planning mitigation measures.

Aim

To design an interactive Tableau dashboard that visualizes hourly air quality data from 15 monitoring stations using **Heatmaps** and **Ridgeline Plots** to identify pollution hotspots, abnormal environmental conditions, and pollutant distributions.

Procedure

1. Import the Dataset

1. Open **Tableau Desktop/Public**.
2. Click **Connect** → **Text File** and select **Air_Quality_Data.csv**.
3. Verify the fields and open **Sheet 1**.

2. Create a Heatmap

1. Drag **Hour** to the **Columns** shelf.
2. Drag **Station** to the **Rows** shelf.
3. Drag **PM2.5** (or any pollutant) to **Color**.
4. Select **Square** in the **Marks** card and rename the worksheet as **Heatmap**.

3. Create a Ridgeline Plot

Note: Tableau does **not** have a built-in Ridgeline Plot.

1. Prepare a dataset with density values (or use a Tableau extension).
2. Drag **Station** to **Rows**.
3. Drag **PM2.5** to **Columns**.
4. Create the Ridgeline Plot using the density field or extension and rename the worksheet as **Ridgeline Plot**.

4. Create the Dashboard

1. Click **New Dashboard**.

2. Add the **Heatmap** and **Ridgeline Plot** worksheets.
3. Add filters for **Station** and **Pollutant**.
4. Save the dashboard.

Explanation

Heatmap

A heatmap displays pollution intensity across monitoring stations and hourly time periods using color gradients. Darker colors represent higher pollution levels, making it easy to identify pollution hotspots and peak pollution hours.

Ridgeline Plot

A ridgeline plot compares the distribution of pollutant levels across multiple monitoring stations. It shows the shape, spread, and concentration of pollution values, helping identify abnormal environmental conditions and differences between stations.

Output



Result

The Tableau dashboard successfully visualizes air quality data using Heatmaps and Ridgeline Plots. It enables environmental authorities to identify pollution hotspots, compare pollutant distributions across monitoring stations, detect pollution spikes, and support pollution mitigation planning.

3. A university tracks student academic performance across multiple departments over five academic years.

Task: Design a visualization dashboard that:

- **Uses Line Charts to show performance trends over time.**
- **Incorporates Scatter Plots to analyze the relationship between attendance and grades.**
- **Uses Facet Grids to compare departmental performance simultaneously.**

Objective: Enable academic administrators to identify performance trends and factors affecting student success.

Aim

To design an interactive Tableau dashboard that visualizes student academic performance across multiple departments over five academic years using **Line Charts, Scatter Plots, and Facet Grids**.

Procedure

1. Import the Dataset

1. Open **Tableau Desktop/Public**.
2. Click **Connect** → **Text File** and select the student dataset.
3. Verify the fields and click **Sheet 1** to start creating visualizations.

2. Create a Line Chart

1. Drag **Year** to **Columns**.
2. Drag **Grade** to **Rows** and change it to **Average (AVG)**.
3. Select **Line** from the **Marks** card.
4. Rename the worksheet as **Performance Trend**.

3. Create a Scatter Plot

1. Open a new worksheet.
2. Drag **Attendance** to **Columns** and **Grade** to **Rows**.
3. Drag **Department** to **Color** and **Student_ID** to **Detail**.
4. Add a **Trend Line** (optional) and rename the sheet.

4. Create a Facet Grid

1. Open another worksheet.
2. Drag **Year** to **Columns** and **Grade** to **Rows**.
3. Drag **Department** to **Rows** to create separate panels.
4. Select **Line** as the chart type and rename the sheet.

5. Create the Dashboard

1. Click **New Dashboard**.
2. Drag the three worksheets onto the dashboard.
3. Add **Year** and **Department** as filters.
4. Save the dashboard.

Explanation

Line Chart

A line chart shows how the average student grades change over five academic years. It helps identify increasing or decreasing performance trends over time. This visualization supports long-term academic performance analysis.

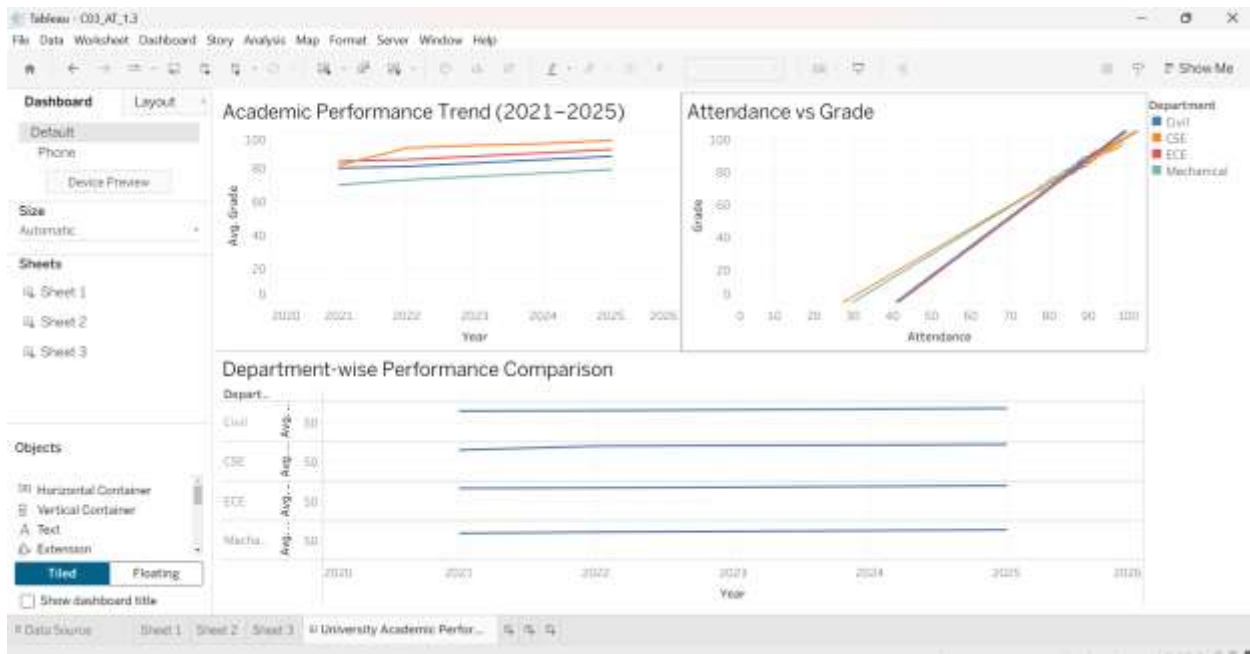
Scatter Plot

A scatter plot displays the relationship between student attendance and grades. Each point represents an individual student, making it easy to observe correlations. It helps determine whether higher attendance is associated with better academic performance.

Facet Grid

A facet grid divides the data into separate panels for each department. This allows easy comparison of academic performance across multiple departments at the same time. It helps identify departments with consistently high or low performance.

Output



Result

The Tableau dashboard successfully visualizes academic performance trends, attendance versus grades, and departmental comparisons. It enables administrators to identify performance trends, compare departments, and make informed academic decisions.

4. A logistics company records delivery times for thousands of shipments using Road, Rail, Air, and Sea transportation modes.

Task: Create a visualization design that:

- **Uses Histograms and ECDFs to compare delivery-time distributions across transportation modes.**
- **Employs Q-Q Plots to assess whether delivery times follow a theoretical normal distribution.**
- **Highlights delays, variability, and outliers in delivery performance.**

Objective: Support operational teams in optimizing transportation strategies and improving delivery reliability.

Aim

To design an interactive Tableau dashboard that analyzes delivery times across Road, Rail, Air, and Sea transportation modes using Histograms, ECDFs, and Q-Q Plots to identify delays, variability, and delivery performance.

Procedure

1. Import the Dataset

1. Open **Tableau Desktop/Public**.
2. Click **Connect** → **Text File** and select the logistics dataset.
3. Verify the fields and open **Sheet 1**.

2. Create a Histogram

1. Create bins for the **Delivery_Time** field.
2. Drag **Delivery_Time (bin)** to **Columns**.
3. Drag **Number of Records** to **Rows**.
4. Drag **Transport_Mode** to **Color**.

3. Create an ECDF Plot

1. Create a calculated field for the cumulative percentage (ECDF).

2. Drag **Delivery_Time** to **Columns**.
3. Drag the **ECDF calculated field** to **Rows**.
4. Drag **Transport_Mode** to **Color**.

4. Create a Q-Q Plot

1. Create calculated fields for **Theoretical Quantiles** and **Sample Quantiles** (or use a prepared dataset with these values).
2. Drag **Theoretical Quantiles** to **Columns**.
3. Drag **Sample Quantiles** to **Rows**.
4. Add a reference line to compare normality.

5. Create Dashboard

1. Create a new dashboard.
2. Add the Histogram, ECDF, and Q-Q Plot.
3. Add a filter for **Transport_Mode**.
4. Save the dashboard.

Explanation

Histogram

A histogram shows the frequency distribution of delivery times for each transportation mode. It helps identify common delivery durations and detect delays. The graph is useful for comparing delivery performance across Road, Rail, Air, and Sea.

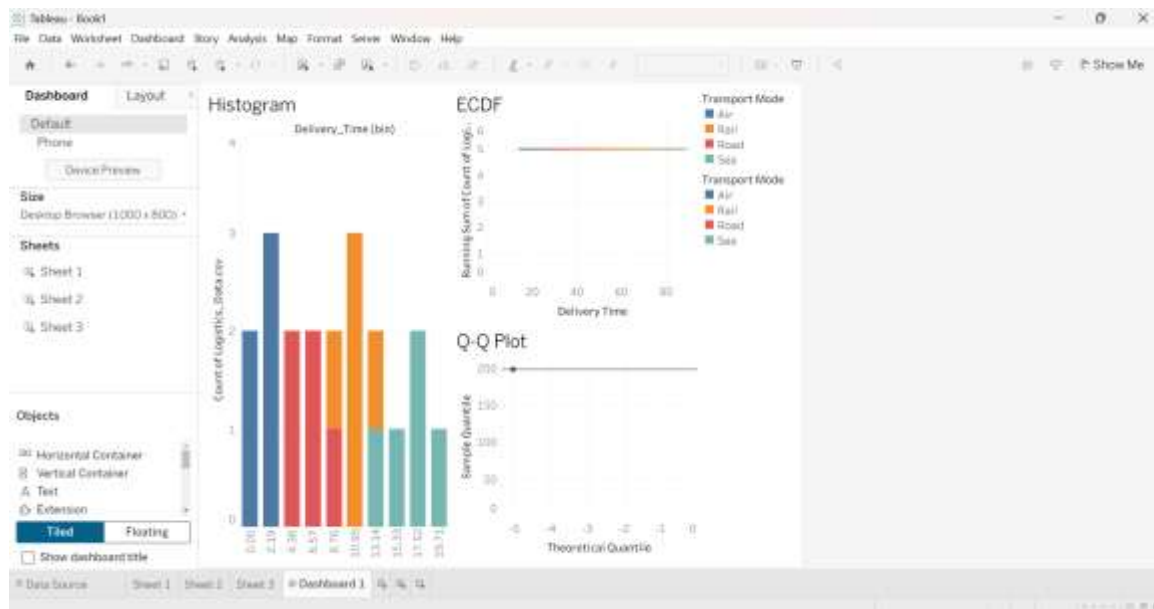
ECDF (Empirical Cumulative Distribution Function)

The ECDF displays the cumulative percentage of deliveries completed within a given time. It allows easy comparison of delivery efficiency among transportation modes. Faster modes will reach higher cumulative percentages more quickly.

Q-Q Plot

A Q-Q plot compares the observed delivery times with a theoretical normal distribution. If the points lie close to the reference line, the data is approximately normally distributed. Deviations from the line indicate skewness, outliers, or unusual delivery patterns.

Output



Result

The Tableau dashboard successfully visualizes delivery-time distributions using Histograms, ECDFs, and Q-Q Plots. It enables comparison of transportation modes, identifies delays and outliers, evaluates the normality of delivery times, and supports logistics teams in improving transportation efficiency and delivery reliability.